E-Commerce Customer Lifetime Value and Cohort Retention Analysis

Executive summary | Online Retail II dataset (Dec 2009 to Dec 2011) | Prepared by: Suvarnalakshmi D

Objective

Identify high-value customer segments, measure how well customers are retained over time, and recommend where to allocate the retention marketing budget.

Data and method

- 1,067,371 transactions were loaded into PostgreSQL. About 25% of rows were removed: no customer ID (23% of all rows), cancelled invoices, and zero or negative quantity or price. The clean data has 805,531 rows from 5,878 customers and total revenue of about 17.7M.
- A monthly cohort retention matrix was built in SQL using CTEs, window functions and a self-join.
- Customers were scored on Recency, Frequency and Monetary value (RFM) and grouped into four segments with K-Means clustering (k = 4) in Python.
- Historical CLV (past spend) and predictive CLV (average order value x orders per year x expected lifespan) were calculated for each customer.
- Results are presented in a three-page interactive Power BI dashboard.

Key findings

Segment	Customers	Revenue share	Avg. historical CLV	Avg. predictive CLV
Champions	20.3%	73.9%	10,979	16,469
At-Risk	24.8%	16.4%	1,992	996
Potential Loyalists	21.4%	6.2%	868	868
Hibernating	33.5%	3.6%	324	81

- One in five customers (Champions) generates nearly three quarters of revenue, and each is expected to be worth about 50% more in future.
- Only about 21% of customers return in the month after their first purchase, and about 18% are still active 12 months later.
- Revenue peaks every November (holiday season). December 2011 is a partial month in the data.
- At-Risk customers have not purchased for about 230 days on average, and they have spent about 2.9M historically. Their predictive CLV is half of their historical CLV, which is where revenue is leaking.
- Hibernating customers are a third of the customer base but only 3.6% of revenue, with almost no predicted future value.

Recommendations

Suggested split of the retention marketing budget:

1. **Champions (about 35%):** VIP and loyalty program, early access and personalized offers, to protect the segment that produces most of the revenue.
2. **At-Risk (about 35%):** win-back campaigns such as reminder emails and time-limited offers, since this is the largest recoverable revenue.
3. **Potential Loyalists (about 25%):** nurture towards a second and third purchase with product recommendations in the first 90 days.
4. **Hibernating (about 5%):** low-cost automated emails only, with no heavy spending.
5. Plan campaigns ahead of the October to November peak.

Assumptions and limitations

- Predictive CLV uses assumed remaining lifespans: Champions 3 years, Potential Loyalists 2 years, At-Risk 1 year and Hibernating 0.5 years. It is based on about two years of data and a simple formula, not a probabilistic model.
- Rows without a customer ID were excluded, so the results describe identified customers only.
- Segment names were assigned by interpreting the average RFM values of each K-Means cluster.